SHRI RAMSWAROOP MEMORIAL UNIVERSITY

Simulation of a Voting System using SciLab

A simple binary voting system built with SciLab — collecting, counting, and declaring results based on majority vote.

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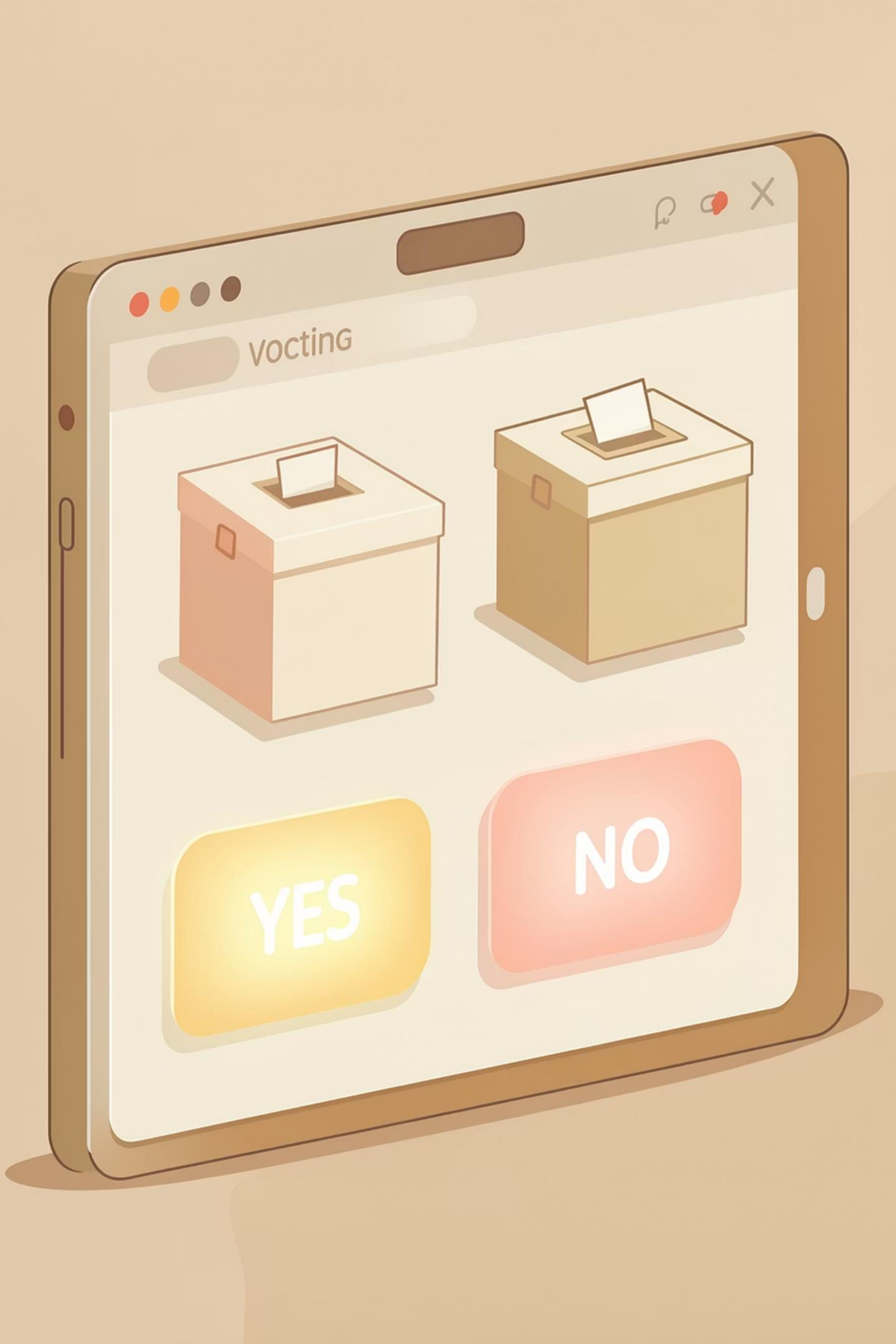


Objective & Introduction

What This Project Does

Designs and implements a simple voting system in SciLab where multiple users cast votes, the system counts them, and declares the winner by majority.

Binary Voting Model

- **1** represents "Yes"
- **0** represents "No"

Demonstrates loops, conditional statements, and user input handling — mirroring real-world voting at a fundamental level.

Methodology

Input Voters
Enter total number of voters

Collect Votes
Record each vote as 1 or 0

Count Votes
Loop through votes to tally Yes/No

Determine Result
Compare counts to decide outcome

Display Output
Show Yes wins, No wins, or Tie

The system walks through a clean five-step pipeline — from voter input to result declaration — making the logic transparent and easy to follow.

Algorithm & SciLab Code

Algorithm

01

Input

Enter number of voters (n)

02

Initialise

yes_votes = 0, no_votes = 0

03

Loop

For each voter, collect and validate vote

04

Compare

Determine Yes / No / Tie

05

Display

Show vote counts and final result

SciLab Code

```
clc; clear;
disp("===== SIMPLE VOTING SYSTEM =====");
n = input("Enter number of voters: ");
yes_votes = 0; no_votes = 0;

for i = 1:n
 vote = input("Enter vote (1=Yes, 0=No): ");
 if vote == 1 then
 yes_votes = yes_votes + 1;
 elseif vote == 0 then
 no_votes = no_votes + 1;
 else
 disp("Invalid vote! Skipped.");
 end
end

if yes_votes > no_votes then
 disp("Final Result: YES wins");
elseif no_votes > yes_votes then
 disp("Final Result: NO wins");
else
 disp("Final Result: TIE");
end
```

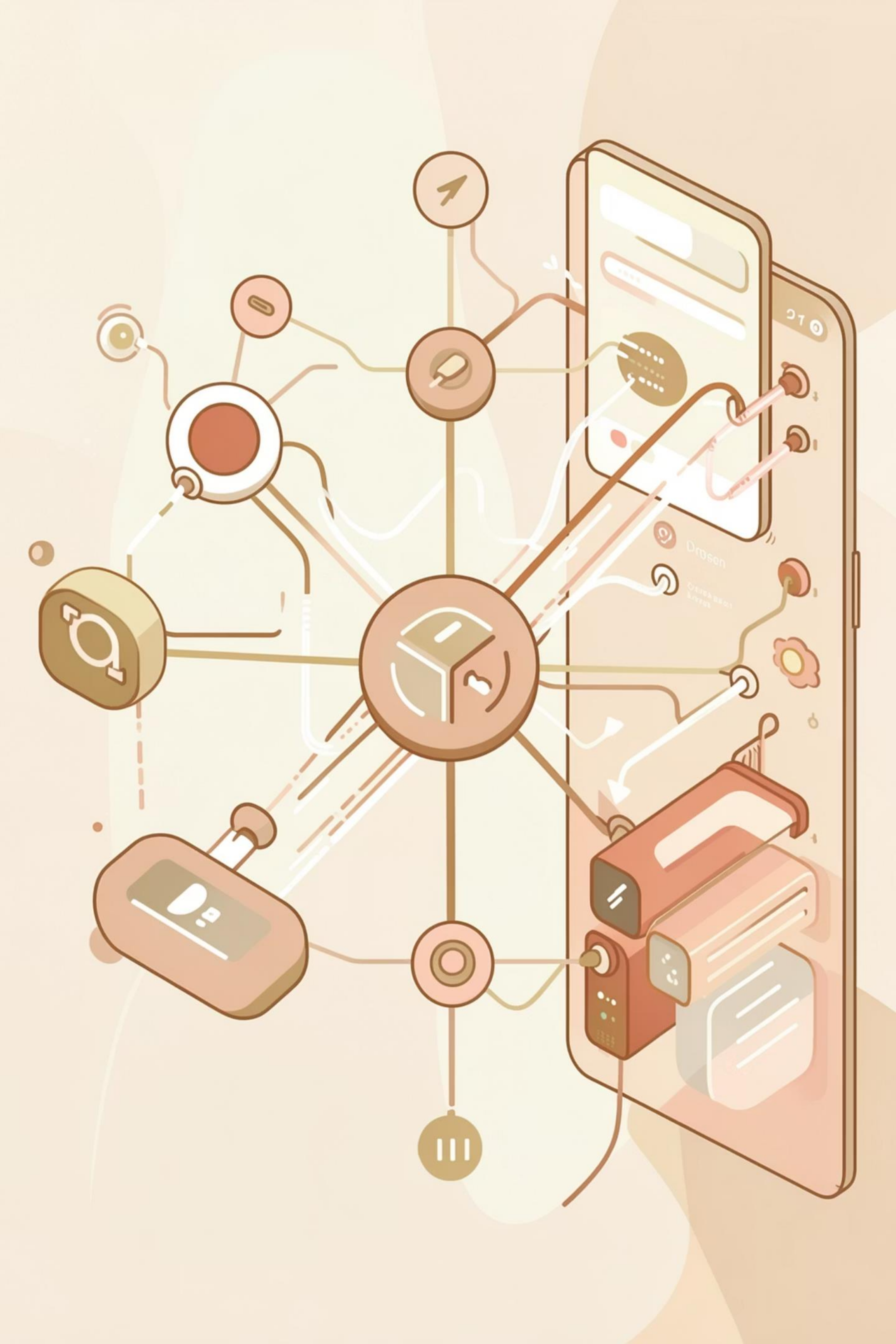
Sample Output

Example Run (5 Voters)

```
Enter number of voters: 5

Voter 1 → 1
Voter 2 → 0
Voter 3 → 1
Voter 4 → 1
Voter 5 → 0

Yes Votes: 3
No Votes:  2

Final Result: YES wins
```

3	2	5
Yes Votes	**No Votes**	**Total Voters**

Final Result: **YES wins** by majority



Conclusion & Future Scope

Advantages & Limitations

Strengths

Simple, efficient, handles multiple voters, demonstrates real-world logic

⚠️ Limitations

Binary only, no voter authentication, no data storage, not production-ready

Future Scope

- Support for multiple candidates
- Password protection & authentication
- GUI interface development
- File-based result storage
- Online voting system